

VulNerd

The nerdy classmate who actually wants you to pass.

What is VulNerd?

VulNerd is a web based tool that reads vulnerability scan reports and turns them into something an actual human can understand. You upload a Nessus CSV, and instead of getting back a 400 row spreadsheet full of CVE codes and severity numbers, you get a report card. A grade. Plain English explanations. A compliance dashboard for your industry. And real numbers on what fixing things costs versus what ignoring them costs.

It was built for the people that traditional cybersecurity tools forgot about. Small business owners who cannot afford a dedicated security team. Students learning vulnerability analysis who need a way to bridge the gap between theory and practice. Anyone who has ever opened a scan report and immediately felt like they were in over their head.

The goal is not to replace a security analyst. The goal is to make sure you do not need to hire one just to understand your own network.

The Problem It Solves

Cybersecurity was never built with small businesses in mind. The tools exist, the scanners exist, the frameworks exist, but all of it assumes you have a dedicated security team sitting behind it making sense of the output. Most small businesses do not have that. What they have is an owner wearing six hats, a Nessus export full of CVE codes, and no idea which one is about to cost them everything.

VulNerd sits in that gap. It does not try to be the SOC team. It tries to be the one person in the room who can read the scan, tell you what actually matters, and walk you through the fix in a way that makes sense.

How It Works

VulNerd runs your scan through five stages before anything reaches your screen.

1. Collect

You upload a Nessus CSV file through the web interface. Drag and drop, or click to browse. That is the entire user effort required. VulNerd handles everything after that.

2. Parse

A custom Python engine reads the raw Nessus export, cleans it, normalizes the data, and structures it into something that can be analyzed. Nessus exports are not exactly friendly even to other programs, so this stage does the heavy lifting of turning raw scanner output into usable input.

3. Analyze

Google Gemini AI reads the cleaned findings and writes a full report card. It grades your network like a teacher grading an exam, identifies your biggest risks, explains what each vulnerability actually means in plain English, and gives you specific, prioritized next steps. This is where VulNerd earns its name. It is not just showing you data. It is teaching you what the data means.

4. Comply

You tell VulNerd what kind of business you run. Healthcare, retail, government, education, or general. It maps your findings to the compliance framework that actually matters for your industry, HIPAA, PCI-DSS, NIST 800-53, FERPA, or CIS Controls. Then it builds a compliance dashboard that shows your score across key framework attributes, explains the framework in plain English, tells a real world story of a similar business that got burned by the same gaps, and gives you an honest cost comparison between fixing the problem now and dealing with a breach later.

5. Deliver

Everything displays in your browser as soon as it is ready. You can read the full report card on the left, see the compliance dashboard on the right, chat with VulNerd directly through a built in AI assistant that has context on your specific scan, and email a polished PDF report card straight to your boss, your IT team, or yourself.

What Is Inside the Report Card

- **Executive Summary.** The one paragraph version. What is the state of your network and what should you care about most.
- **Overall Grade.** A letter grade from A to F that reflects the real risk level of your scan.
- **Key Findings.** The specific vulnerabilities that are pulling your grade down, explained like a human being wrote them.
- **Contributing Factors.** The patterns and underlying issues behind the findings.
- **Recommendations.** Concrete, prioritized next steps. Not a generic checklist.
- **Threat Context.** How these vulnerabilities map to real world attacker behavior.
- **Who to Call.** Guidance on who in your organization should actually handle each item.
- **Instructor's Note.** A personal message from VulNerd, written in the voice of the nerdy classmate who wants you to succeed.

The Compliance Dashboard

Next to your report card, VulNerd builds a side panel dedicated to compliance for your specific industry. It shows five framework attributes scored from 0 to 100, with color coded bars so you can see at a glance where you stand. It explains what the framework is and why it matters in language that does not assume you already know. It tells you a real story of a business like yours that paid the price for gaps like these. And it gives you a direct cost comparison. What it would take to fix things now, versus what a breach would actually cost.

The compliance dashboard is the part most business owners never get to see because traditional tools assume you already have a compliance officer. VulNerd assumes you do not, and builds the whole thing for you.

Ask VulNerd Anything

There is a floating chat button in the corner of every results screen. Click it and you get direct access to VulNerd as a conversational assistant. It knows the details of the scan you just uploaded and can answer follow up questions, explain any finding in more detail, walk you through a specific fix, or clarify anything in the report that did not land the first time. It is the classmate staying late to help you study.

Built With

- **Python.** Core language for the web server, scan parser, and orchestration logic.
- **Google Gemini AI.** Powers the report card generation, compliance analysis, and chat assistant.
- **Nessus Essentials.** The vulnerability scanner whose output VulNerd was built to read.
- **DVWA.** The vulnerable test environment used for development and validation.
- **Docker.** Containerized lab environment for repeatable testing.
- **RealVNC.** Remote access to the scan environment during development.
- **ReportLab.** PDF generation for emailable report cards.
- **Python stdlib HTTP server.** Runs the entire web app without needing Flask or any external framework.
- **HTML, CSS, JavaScript.** Custom dark themed single page interface with animated compliance charts and floating chat widget.
- **SMTP.** Delivers finished report cards directly to stakeholders' inboxes.

Who It Is For

- **Small business owners.** Who need to know if their network is safe without hiring a security team to tell them.
- **Cybersecurity students.** Who need a real tool to practice vulnerability analysis and learn how to translate raw findings into actionable reports.
- **IT generalists.** Who inherited security responsibility without security training and need a clear path forward.
- **Anyone who has ever opened a Nessus export and closed it again.** Because they had no idea what to do with it.

Why It Matters

The gap between enterprise grade cybersecurity and the rest of the world is massive. Breaches do not just happen to companies that ignore security. They happen to companies that tried, got overwhelmed, and gave up because the tools were not built for them. VulNerd is an attempt to close that gap. Not by dumbing security down, but by translating it. By giving people real answers, real explanations, and real next steps instead of a spreadsheet and a prayer.

Security does not have to feel impossible. It just needs someone to explain it like you are in this together. That is what VulNerd is.